BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

ELECTROMAGNETIC FIELD THEORY

Exam Date: Thursday, 5 Nov 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2024	Electromagnetic Theory	`104404`	B.Tech 4th Semester Examination, 2024
2023	Electromagnetic Theory	`104404`	End Semester Examination - 2023
2020	Electromagnetic Field Theory	`100308`	B.Tech 3rd Semester Exam., 2020 (New Course)

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Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

ELECTROMAGNETIC THEORY

Time: 03 Hours | **Code: 104404** | **Full Marks: 70**

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[14]**

1. The divergence of the vector field is
   - i. Vector
   - ii. Scalar
   - iii. Can't be both
   - iv. None

2. The unit of electric field intensity
   - i. Volt
   - ii. Volt per metre
   - iii. Tesla
   - iv. Metre

3. In Gauss's Law the electric field is related to
   - i. The charge density
   - ii. The potential difference
   - iii. The charge enclosed
   - iv. The capacitance

4. The electric field at a point due to an infinite sheet of charge is:
   - i. $\frac{\sigma}{\epsilon_0}$
   - ii. $\frac{\sigma}{2\epsilon_0}$
   - iii. $\frac{2\epsilon_0}{\sigma}$
   - iv. 0

5. The force on a charged particle moving in a magnetic field is maximum when the angle between the velocity and magnetic field is:
   - i. $0°$
   - ii. $45°$
   - iii. $90°$
   - iv. $180°$

6. The energy density in magnetic field is given by
   - i. $\frac{\mu H^2}{2}$
   - ii. $\frac{H^2}{2\mu}$
   - iii. 2
   - iv. None

7. The dot product of the vectors $3i - 2j + 5k$ and $-i + 3j + 2k$ is,
   - i. -1
   - ii. 1
   - iii. 0
   - iv. -3i

8. The Pointing vector P is
   - i. $E \times H$
   - ii. $E \cdot H$
   - iii. $\frac{1}{2}E \times H$
   - iv. $(E \times H)^2$

9. The Maxwell's equation $\nabla \cdot B = 0$ is due to
   - i. Monopole does not exit
   - ii. Coulomb's Law
   - iii. Existence of monopoles
   - iv. Ampere Circuital Law

10. Curl of gradient of a vector is
    - i. 0
    - ii. Null Vector
    - iii. Unity
    - iv. Depends on the constants of the vector

Q.2 Solve both questions : **[7+7=14]**

1. Write the differential elements (dl, da, dv) in both Cartesian, cylindrical co-ordinate system.
2. State divergence theorem. What will be divergence to position vector?

Q.3 Solve both questions : **[7+7=14]**

1. Given the two points, $C(-3,2,1)$ and $D(r=5,\theta=20°,\varphi=-70°)$. find: (i) The spherical coordinates of C (ii) The Cartesian coordinates of D
2. Find the divergence of $\vec{A}=3x^2a_x+5x^2y^2a_y+xyz^3a_z$ where a_x, a_y and a_z are unit vectors in cartesian coordinates at point (1,1,1)

Q.4 Solve both questions : **[7+7=14]**

1. State and derive Coulomb's law. Write coulomb's law in vector forms.
2. Derive an expression for intensity of electric field at a point distant r from a point charge.

Q.5 Solve both questions : **[7+7=14]**

1. An infinite long line charge of uniform density ρ_L C/cm is situated along the z-axis. Obtain electric field intensity due to this charge using Gauss's law.
2. Derive energy density in electrostatic field.

Q.6 Solve both questions : **[7+7=14]**

1. Four 3pC charges are at the corners of a 1-m square. The two charges at the left side of the square are positive. The two charges on the right side are negative. Find the field E at the centre of the square, $\epsilon_r=1$
2. What do you understand by capacitance of a capacitor? Deduce and expression for the capacitance of a parallel plate capacitor. How will it be modified when the gaps between the plates is filled with a dielectric?

Q.7 Solve both questions : **[7+7=14]**

1. Show that the Faraday's law of electromagnetic induction can be expressed as $\nabla\times E=-\partial B/\partial t$. Write down its integral form.
2. Explain the concept of displacement current and show how it led to the modification of the Ampere's law.

Q.8 Solve both questions : **[7+7=14]**

1. Discuss reflection of plane electromagnetic wave incident normally on a perfect dielectric and obtain expressions for the two reflection coefficients of electric and magnetic fields.
2. Define Poynting vector. Mention any two properties of uniform plane wave.

Q.9 Write short notes on any two of the following:- **[14]**

a. Biot-Savart's Law

b. Magnetic field boundary condition for isotropic medium

c. Skin Effect

d. Spherical capacitance

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023

ELECTROMAGNETIC THEORY

Time: 03 Hours | Code: 104404 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven question only: [14]

a. What is physical significance of divergence?

b. State Stoke's theorem.

c. How is the unit vectors defined in three co ordinate systems?

d. State Gauss law for electric fields.

e. Define dielectric strength.

f. State Biot-Savarts law.

g. What is magnetic susceptibility?

h. Write down the wave equation for E and H in free space.

i. State Poynting Theorem.

j. Define intrinsic impedance or characteristic impedance.

Q.2 Solve both questions : [7+7=14]

1. Derive the expression for the attenuation constant, phase constant and intrinsic impedance for a uniform plane wave in a good conductor.

2. Calculate the depth of penetration in copper at 10 MHZ. Given the conductivity of copper is $5.8 \times 10^7 S/m$ and its permeability $1.3mH/m$.

Q.3 Solve both questions : [7+7=14]

1. Derive suitable relations for integral and point forms of poynting theorem.

2. Discuss about the plane waves in lossless dielectrics.

Q.4 Solve this question : **[14]**

1. With explanation, derive the Maxwell's equation in differential and integral forms.

Q.5 Solve both questions : **[7+7=14]**

1. An iron ring with a cross sectional area of 3 cm^2 and mean circumference of 15 cm is wound with 250 turns wire carrying a current of 0.3A. The relative permeability of ring is 1500. Calculate the flux established in the ring.
2. Derive the expressions for boundary conditions in magnetic fields.

Q.6 Solve both questions : **[7+7=14]**

1. State and proof gauss law and explain applications of Gauss law.
2. Explain Poisson's and Lapace's equations.

Q.7 Solve both questions : **[7+7=14]**

1. State and proof divergence theorem.
2. Define divergence, gradient, curl in spherical co-ordinate system with mathematical expression

Q.8 Solve both questions : **[7+7=14]**

1. Define and explain Biot-Savart Law.
2. Derive General field relation for time varying electric and magnetic fields using Maxwell's equations.

Q.9 Write short notes on any two of the following:- **[14]**

a. Equipotential Surface.

b. Magnetic Vector potential.

c. UHF Transmission line.

d. Smith Chart.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 3rd Semester Exam., 2020 (New Course)

ELECTROMAGNETIC FIELD THEORY

Time: 3 hours | **Code: 100308** | **Full Marks: 70**

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.
5. Symbols and notations carry their usual meanings.

Q.1 Choose the correct answer of any seven of the following: **[14]**

1. In free space, the Poisson equation becomes
   - i. Maxwell equation
   - ii. Ampere equation
   - iii. Laplace equation
   - iv. steady-state equation

2. Poisson equation can be derived from which of the following equations?
   - i. Point form of Gauss law
   - ii. Integral form of Gauss law
   - iii. Point form of Ampere law
   - iv. Integral form of Ampere law

3. Poynting vector gives the
   - i. rate of energy flow
   - ii. direction of polarization
   - iii. intensity of electric field
   - iv. intensity of magnetic field

4. Using volume integral, which quantity can be calculated?
   - i. Area of cube
   - ii. Area of cuboid
   - iii. Volume of cube
   - iv. Distance of vector

5. Electric flux density in electric field is referred to as
   - i. number of flux lines
   - ii. ratio of flux lines crossing a surface and the surface area
   - iii. direction of flux at a point
   - iv. flux lines per unit area

6. Which of the following correctly states Gauss law?
   - i. Electric flux is equal to charge
   - ii. Electric flux per unit volume is equal to charge
   - iii. Electric field is equal to charge density
   - iv. Electric flux per unit volume is equal to volume charge density

7. Find the power reflected in a transmission line, when the reflection coefficient and input power are 0.45 and 18 W respectively.
   - i. 3.645
   - ii. 6.453
   - iii. 4.563
   - iv. 5.463

8. In a waveguide, which of the following conditions is true always?
   - i. Phase velocity = c
   - ii. Group velocity = c
   - iii. Phase velocity > c
   - iv. Phase velocity < c

9. The phase and group velocities do not depend on which of the following?
   - i. Frequency
   - ii. Wavelength
   - iii. Phase constant
   - iv. Attenuation constant

Q.2 Attempt any two parts of the following: **[7x2=14]**

1. A rectangular waveguide with dimensions of $3\ \text{cm} \times 2\ \text{cm}$ operates at 10 GHz. Find: (i) cut-off frequency (f_c); (ii) cut-off wavelength (λ_c); (iii) guided wavelength (λ_g); (iv) phase constant (β_g).
2. What do you mean by transmission line? Derive an expression for transmission line equations.
3. Determine the expression for average power of Poynting vector.

Q.3 Attempt any two parts of the following: **[7x2=14]**

1. (i) Define quality factor. Give its relation with attenuation factor. (ii) Define reflection coefficient and VSWR. Also write their interrelation.
2. (i) Compare wave impedance and characteristic impedance. (ii) Define tangent loss.
3. Derive the field components when wave is propagating inside a rectangular waveguide with TM mode propagation.

Q.4 Attempt any two parts of the following: **[7x2=14]**

1. Derive an expression for input impedance when transmission line is terminated with any load impedance.
2. What is equipotential surface? Explain Poynting vector and average Poynting vector.
3. State and prove Ampere's work law as $\nabla \times \vec{H} = J$.

Q.5 Attempt any two parts of the following: **[7x2=14]**

1. Derive the Gauss divergence theorem and Stokes' theorem along with their significances.
2. Explain the wave between parallel planes. Derive the expression for the attenuation in parallel plane guide.
3. Derive the expressions for the reflection and refraction of the waves by the perfect dielectric.

Q.6 Attempt any two parts of the following: **[7x2=14]**

1. Find the reflection and transmission coefficient for the interface between air and freshwater $\epsilon + j180$ in the case of perpendicular incidence.
2. Derive the relationship between the following: (i) Standing-wave ratio and magnitude of reflection coefficient. (ii) Standing-wave ratio and the reflection coefficient.
3. (i) Write the condition for a line to be distortionless. (ii) Define the term 'phase velocity'.

Q.7 Attempt any two parts of the following: **[7x2=14]**

1. What is polarization of wave? Discuss the properties of S- and P-polarized light. Explain why P-polarized light is also called as TM-polarized light.

2. Explain the term 'standing-wave ratio' related to transmission line. What will be the values of input impedances when output impedances are (i) open-circuited and (ii) short-circuited?

3. Explain why TEM wave does not propagate in waveguide.

Q.8 Attempt any two parts of the following: **[7x2=14]**

1. A transmission line has a characteristic impedance of 100 ohms and is terminated in a load impedance of $200 + j180$ ohms. Find the voltage reflection coefficient.

2. What is the penetration depth in current penetration in copper at a frequency of 10^4 MHz, if the resistivity is $1.7 \times 10^{-6}\Omega$ cm?

3. What are the satisfactory conditions for low-loss transmission lines?

Q.9 Attempt any two parts of the following: **[7x2=14]**

1. A uniform plane wave propagating in a medium has $E = 2e^{-\alpha z}\sin(10^8 t - \beta z)a_y$. If the medium is characterized by $\epsilon_r = 1$, $\mu_r = 20$ and $\sigma = 3\ \text{mhos/m}$, then find α, β and $\vec{H}$.

2. In a non-magnetic medium $E = 4\sin(2\pi \times 10^7 - 0.8x)a_z\ \text{V/m}$. Find (i) the time-average power carried by the wave; (ii) the total power crossing $100\ \text{cm}^2$ of plane $2x + y = 5$.

3. What is the boundary condition for metal dielectric interface?